

MATH/COSC 4340

**Numerical Methods for
Differential Equations**

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Course Material Overview

1. Ordinary Differential Equations (ODEs)
solves for a scalar function $f(t)$ using initial condition
or a system, or set of equations: $x(t)$, $y(t)$, $z(t)$
2. Boundary Value Problems (BVPs)
incorporates information about f at/along a boundary
3. Numerical Linear Algebra (NLA)
efficiently solves linear systems and computes spectral
decompositions
4. Partial Ordinary Differential Equations (PDEs)
solves a function defined over a D -dimensional space $f(x,t)$ or
 $f(x,y,z,t)$

ODE Applications: *Predict Celestial Motion*

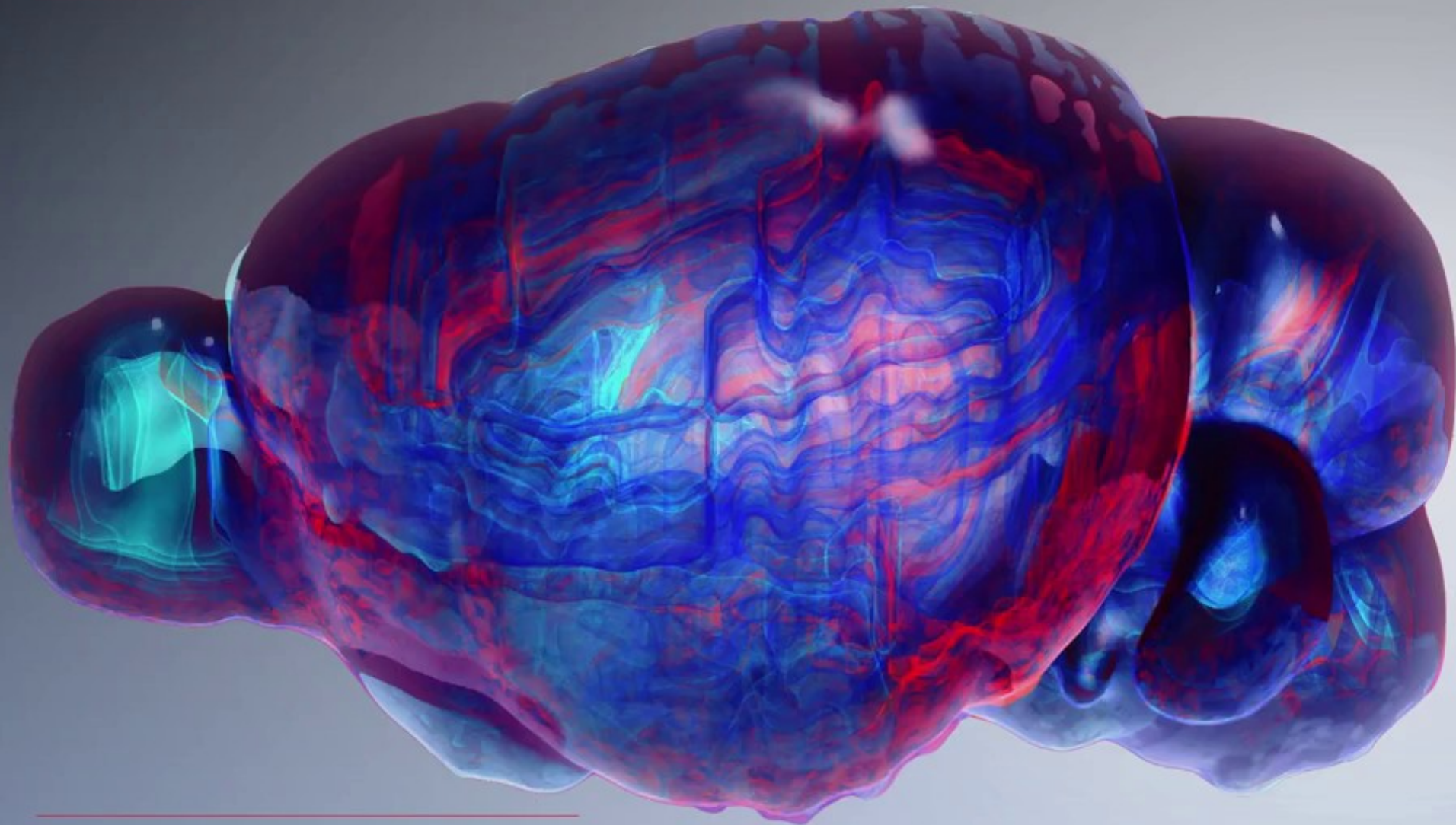


**Solar System with
Unity Physics**



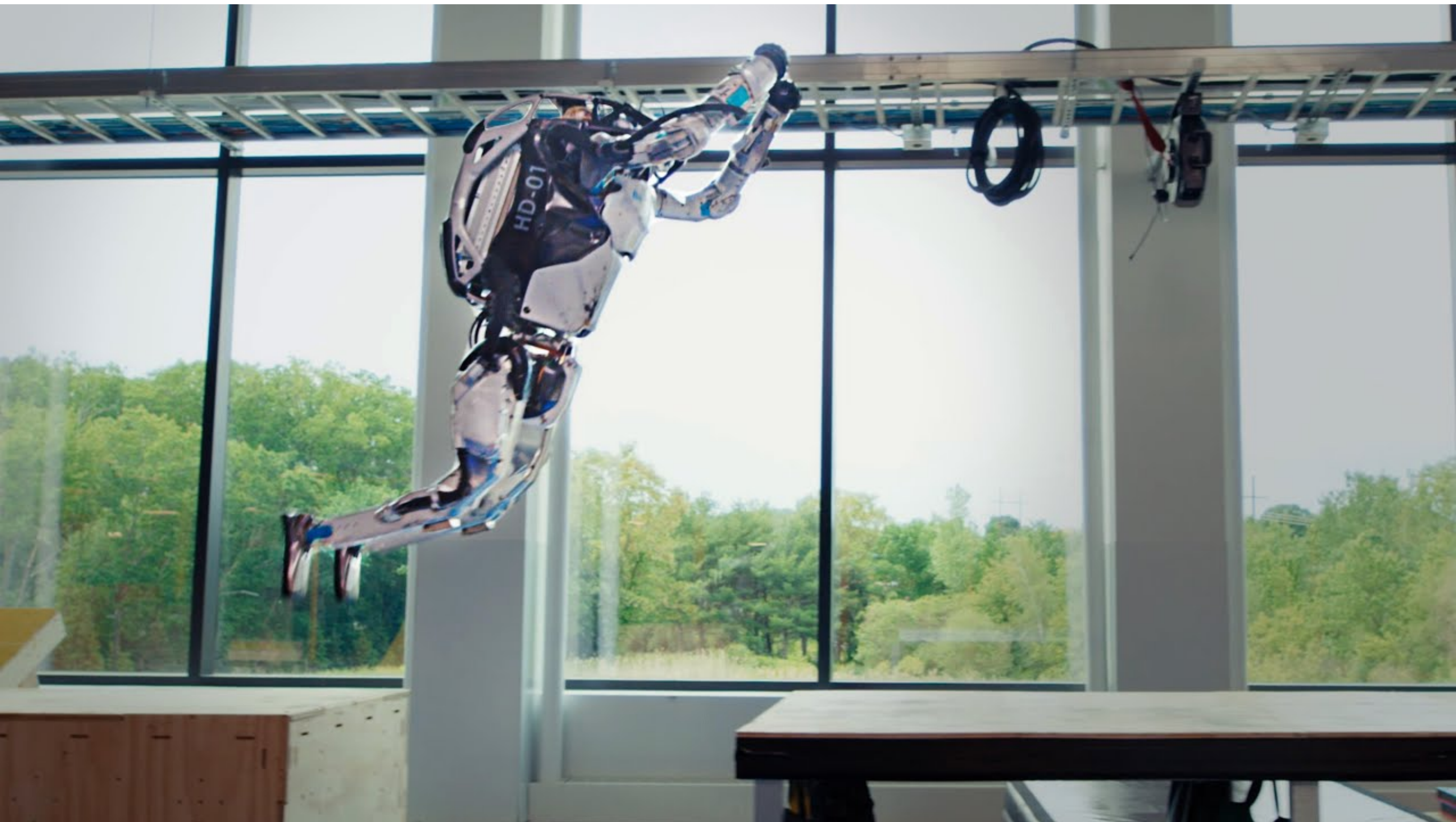
ODE Applications: *Simulate Brain Dynamics*

Blue Brain - 2016
Nicolas Antille



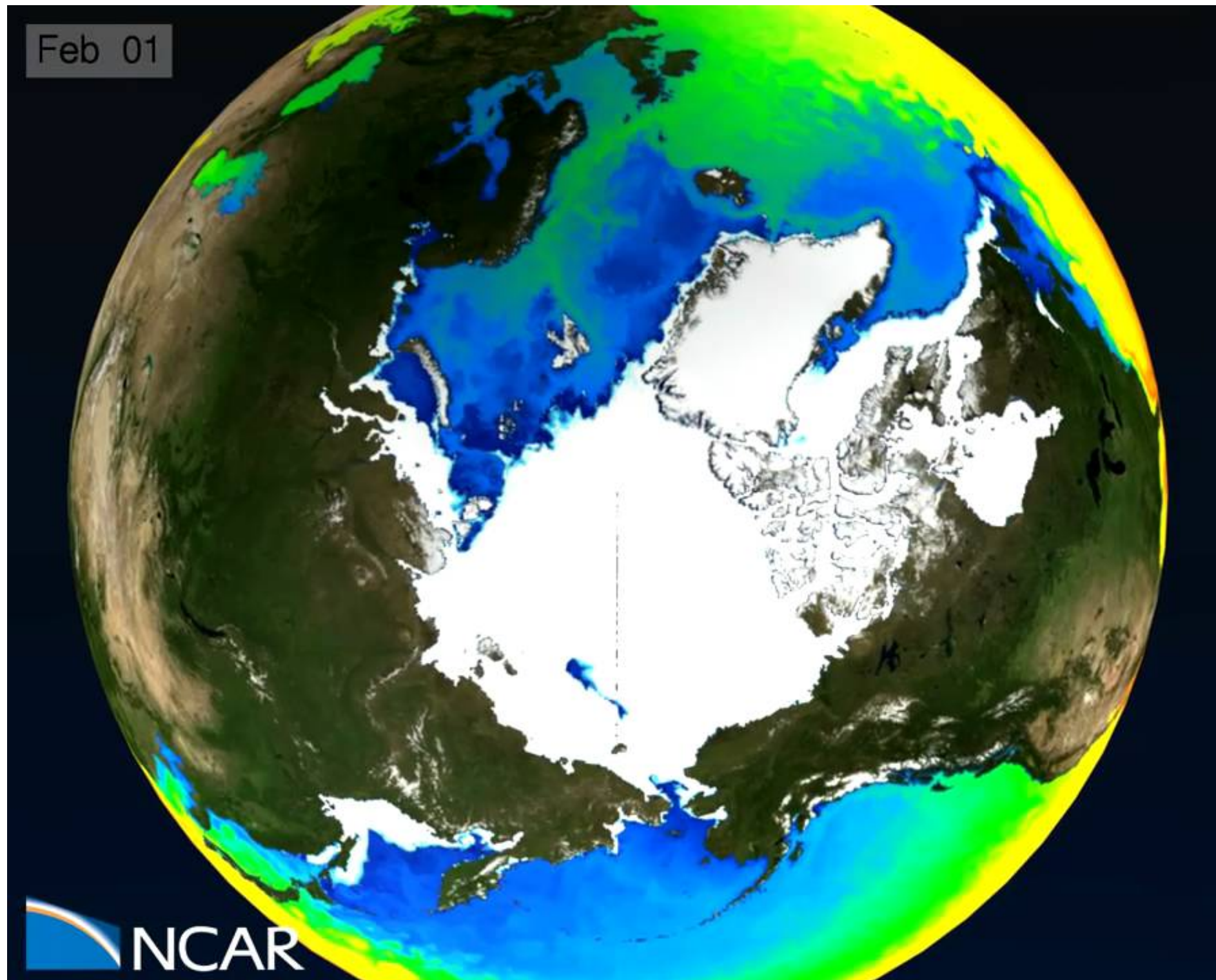
*865 regions in the mouse brain
from the Allen brain atlas*

ODE Applications: *Train/Define Robot Motions*

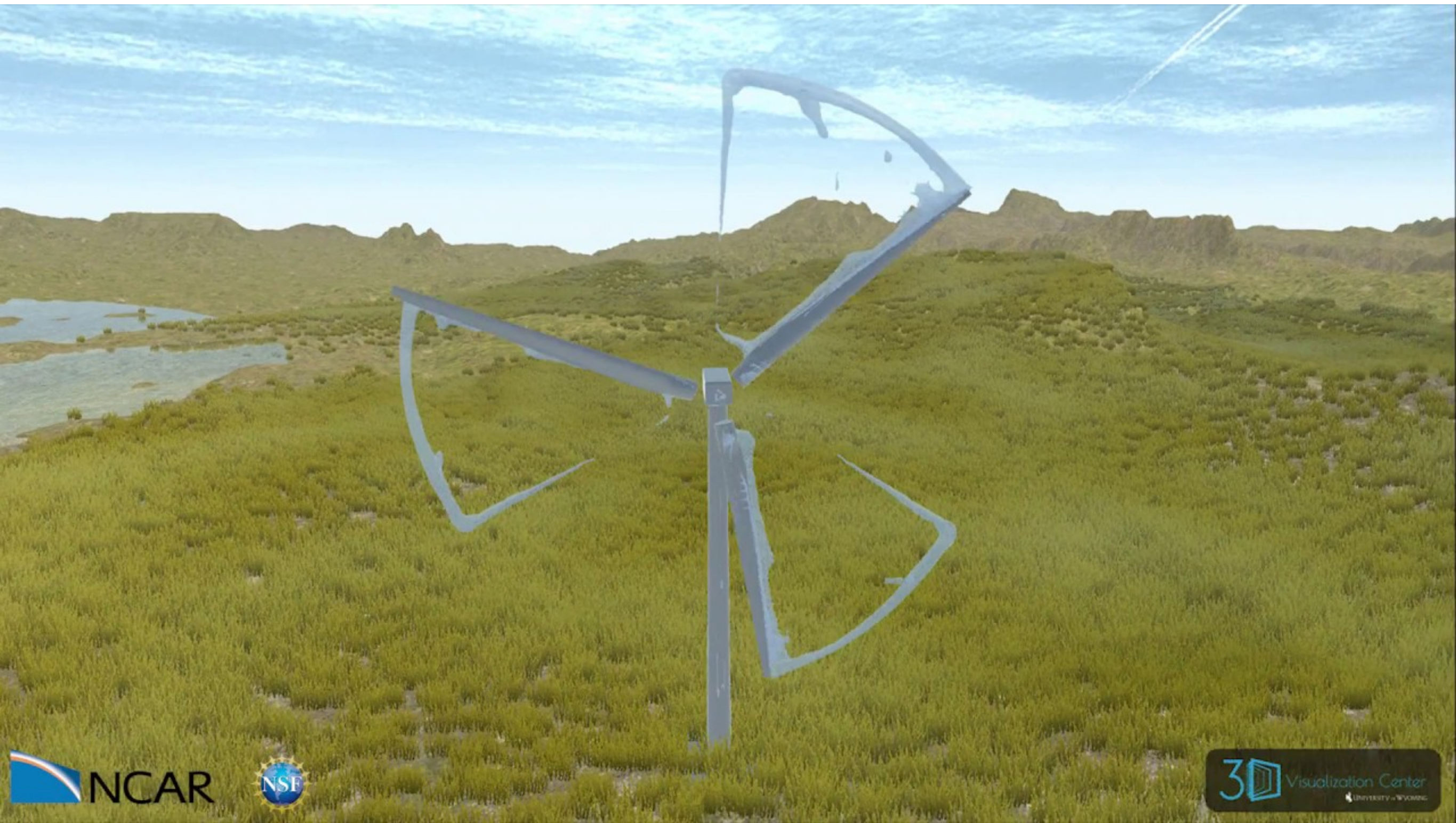


PDE Applications:

Climate Simulation and Prediction



PDE Applications: *Simulate and Optimize Wind Farms*



NCAR Supercomputing Facility in Cheyenne



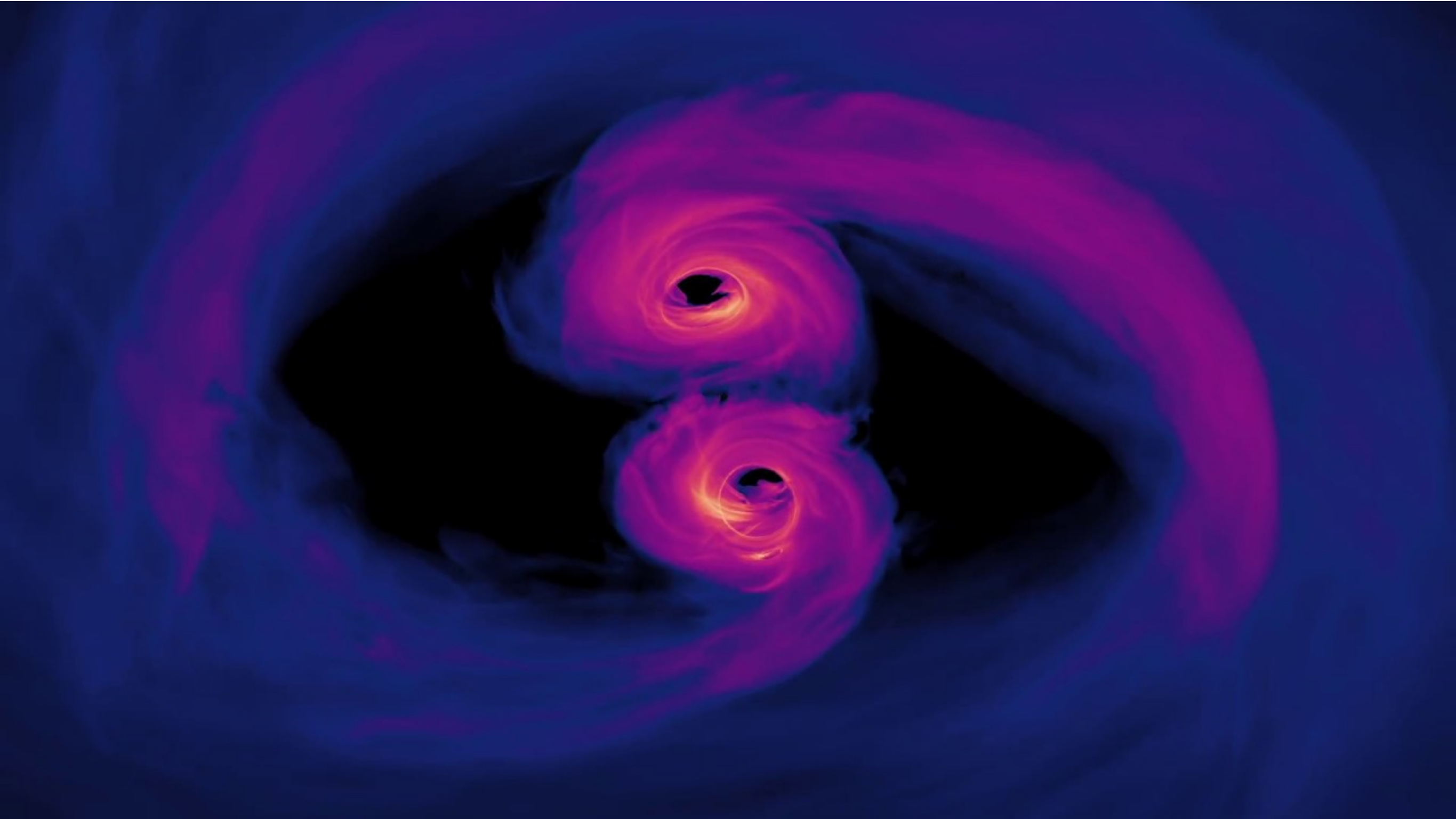
School of Computing Opportunities for Research



SCHOOL OF
COMPUTING

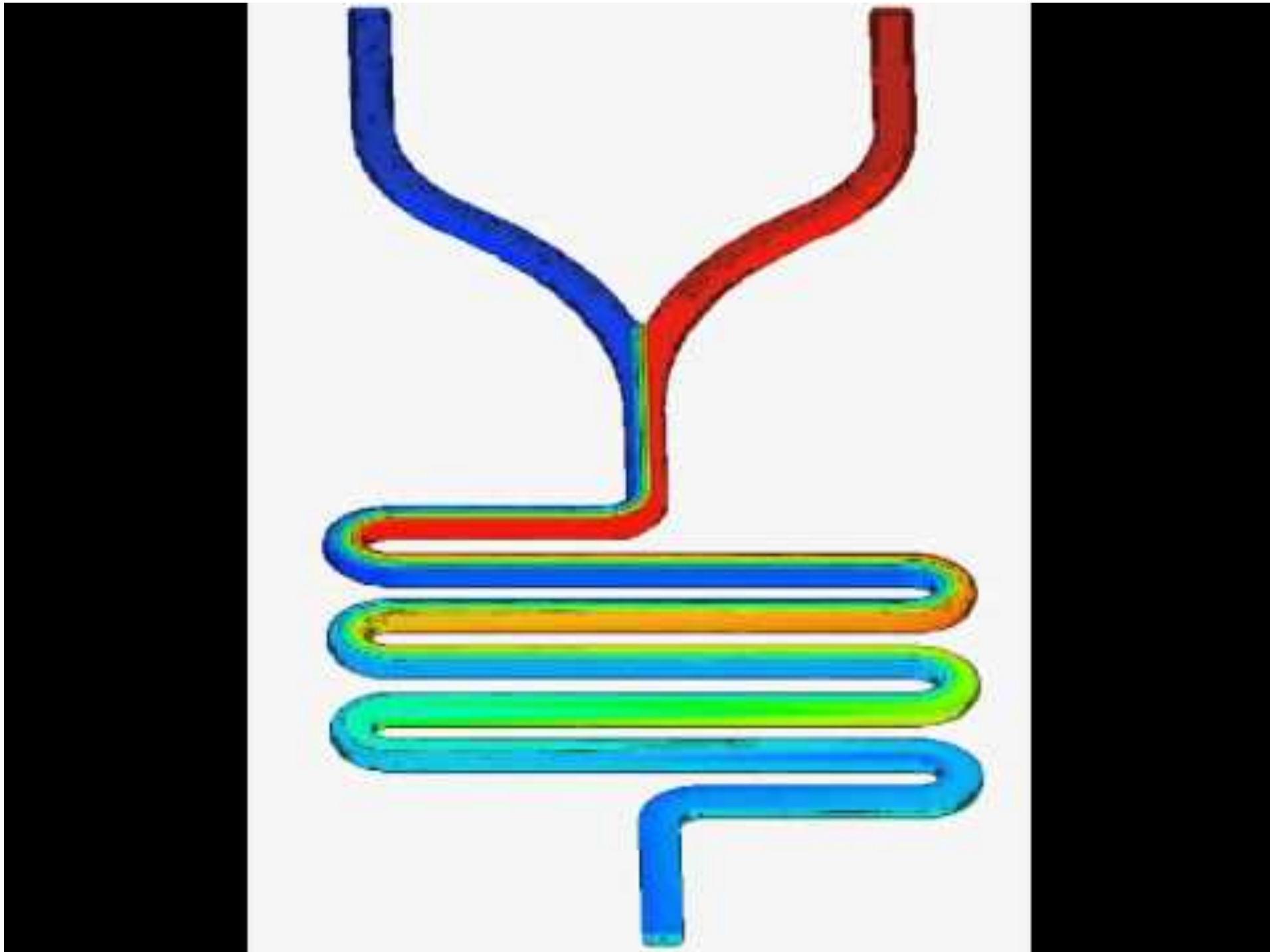
- Many researchers at UW are looking for research assistants with experience writing code in Python, including developing solutions to differential equations.
- Opportunities for paid summer internships and part-time positions during the semester.
- Email me if you are interested: dane.taylor@uwyo.edu

PDE Applications: *Simulate Black Holes*



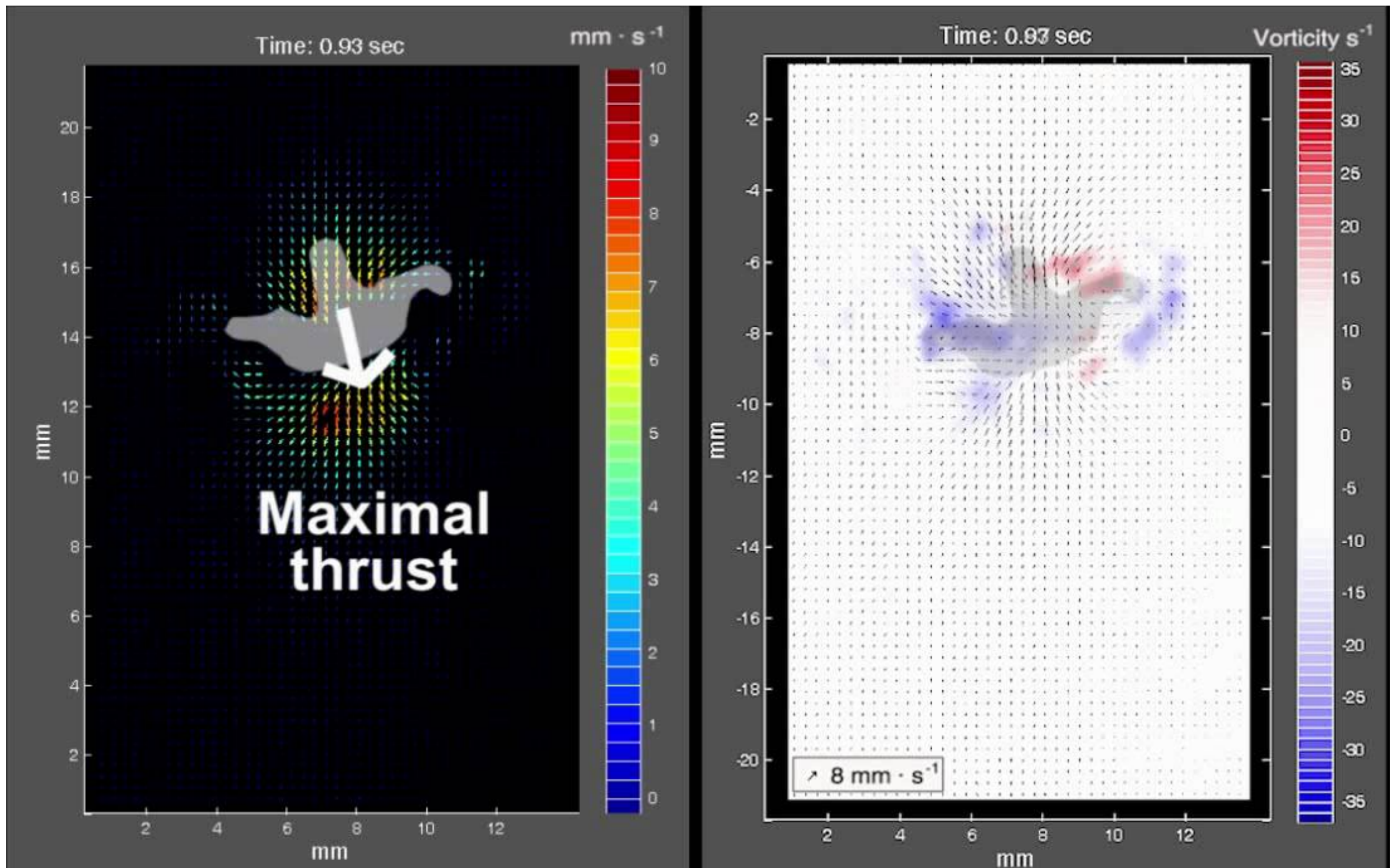
PDE Applications:

Simulate Mixing on a Microfluidic Lab



PDE Applications:

Simulate how Jellyfish Move



PDE Applications:

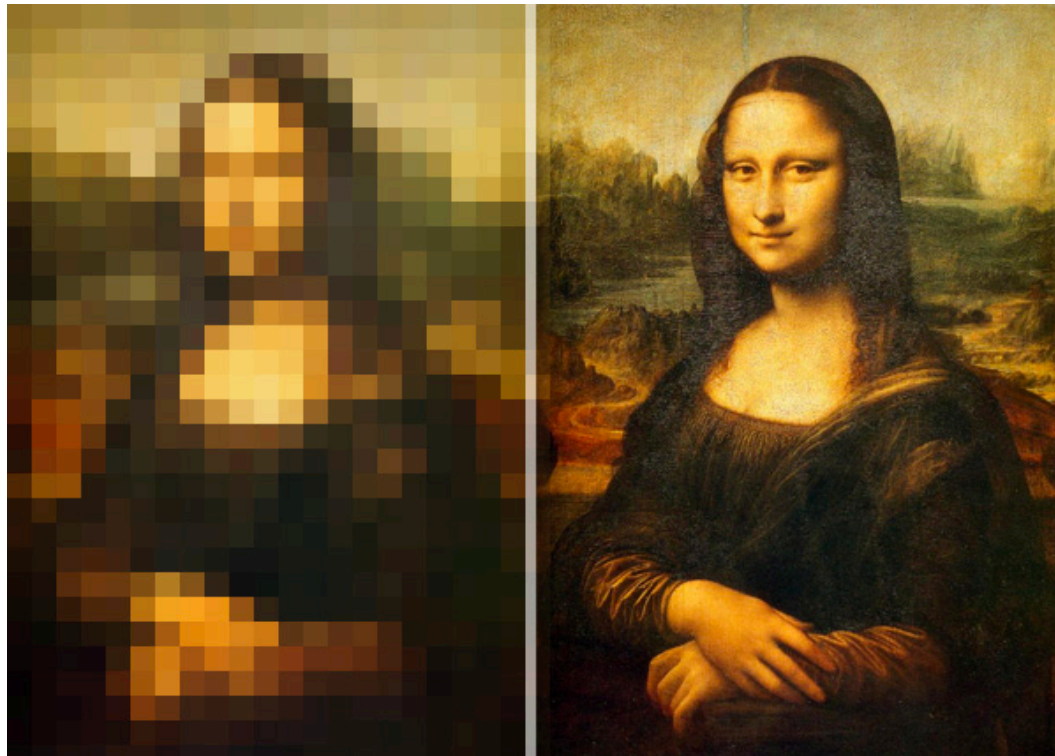
Simulate a Blood Flow in a Heart



Course Goals

1. Understand methods to simulate/solve ODEs and PDEs
2. Develop a strong mathematical understanding of error for these “numerical approximations”
3. Know how to balance approximation error and computational efficiency (accuracy vs. slow code)

high vs low resolution



approximation accuracy

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

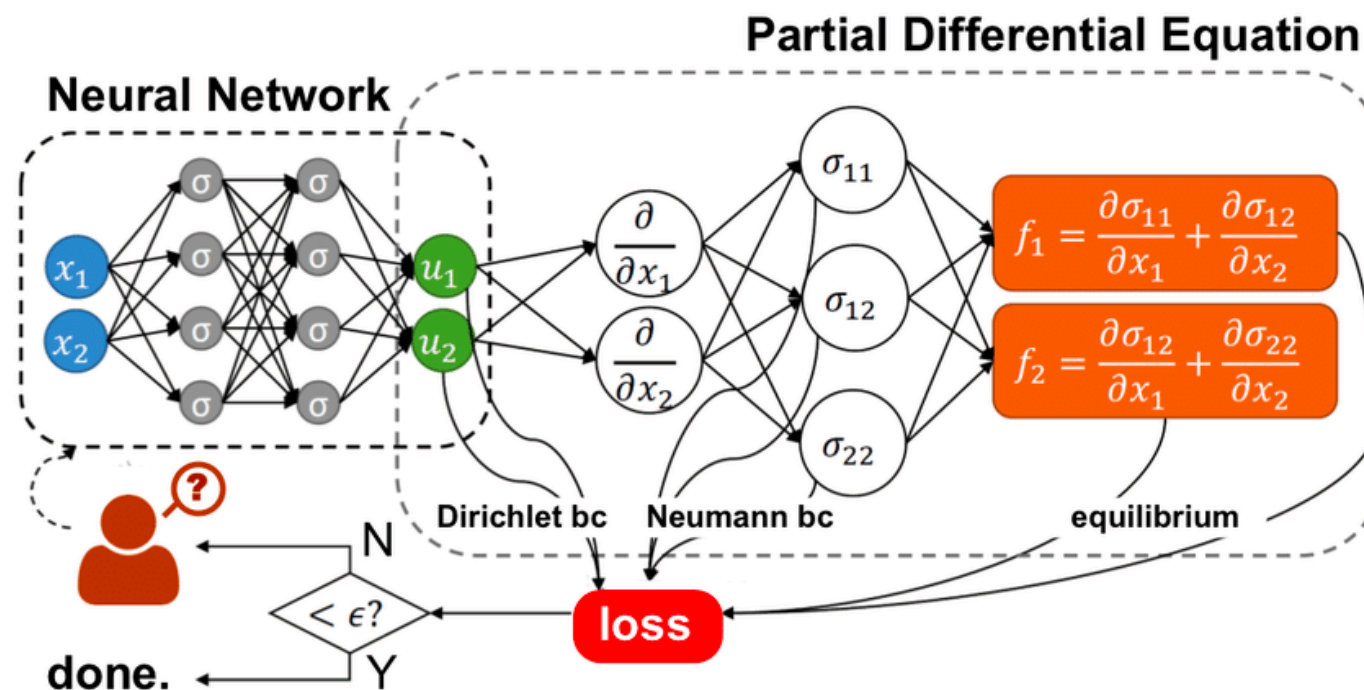
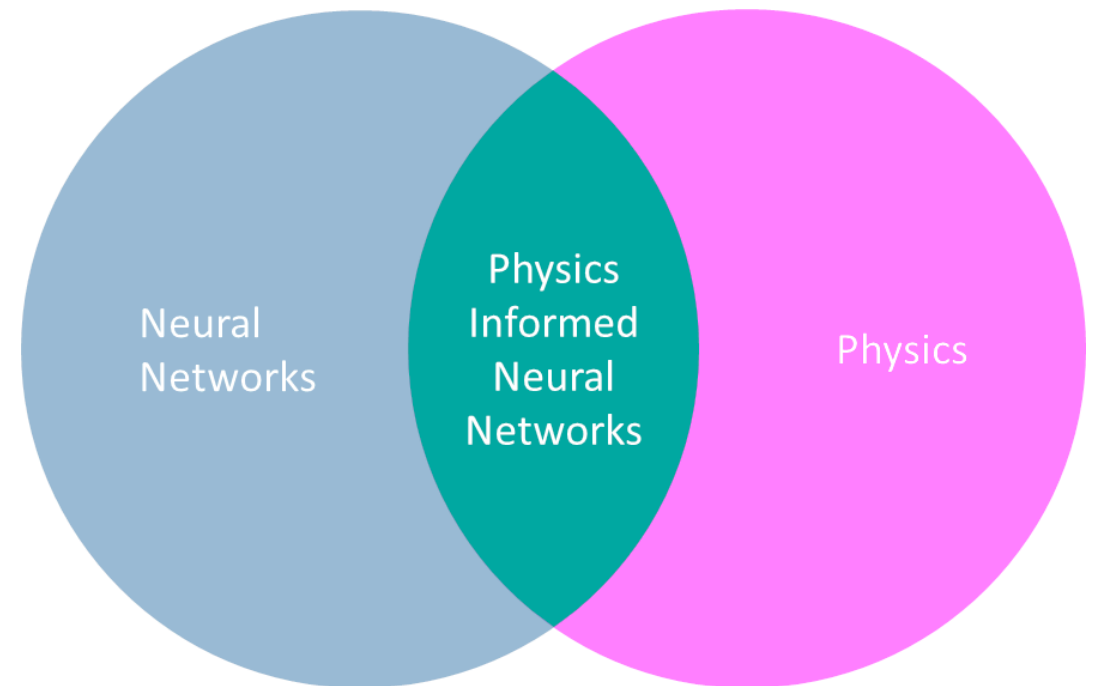
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$$

Course Material Revisited

1. Ordinary Differential Equations (ODEs)
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5. *time permitting*: Other topics may include
random processes and stochastic ODEs
data-driven modeling of ODEs/PDEs

Emerging Applications: *Learning Diff. Eqs. From Data*

- What if we have measured data but we don't know the differential equation?



Emerging Applications: *Neural ODEs and PDEs*

We can design neural networks to “look like” numerical solutions to differential equations

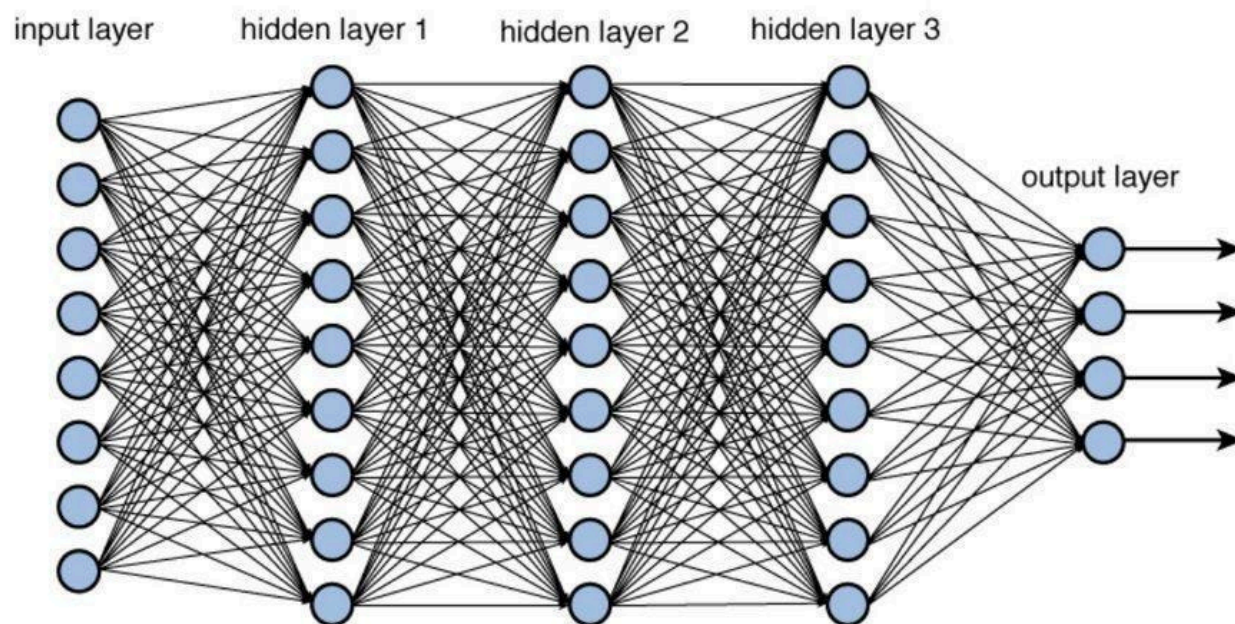
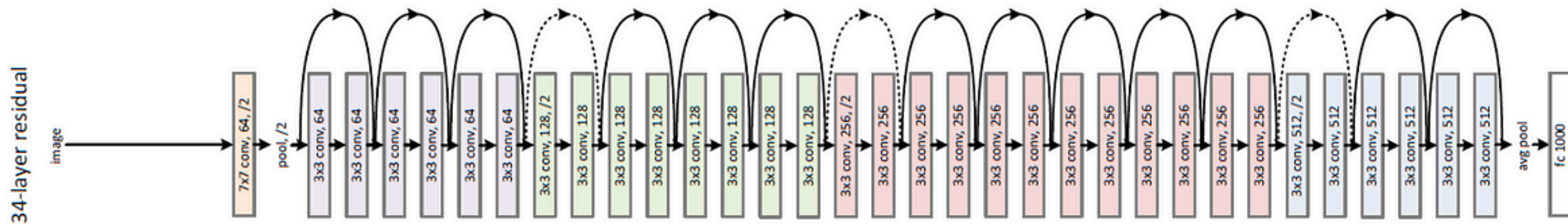
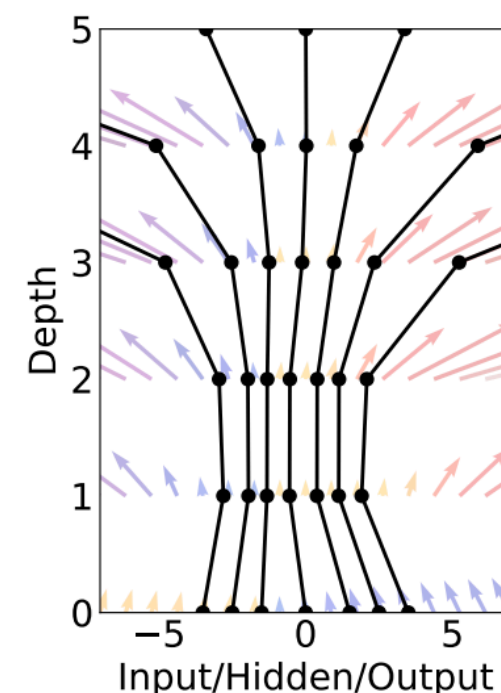
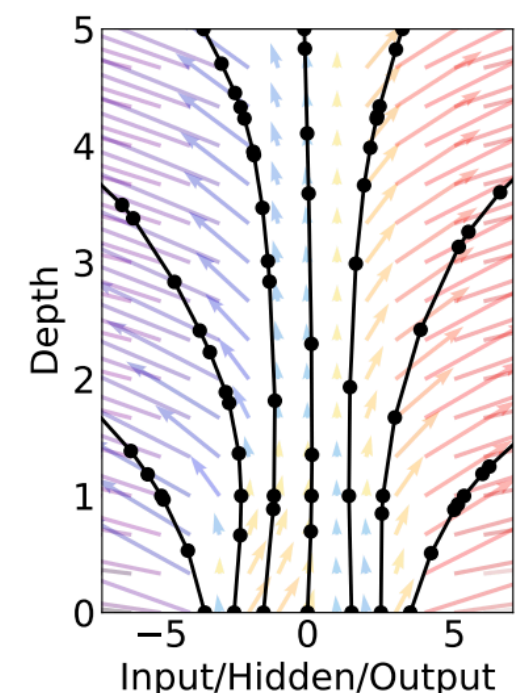


Figure 12.2 Deep network architecture with multiple layers.

Residual Network



ODE Network



Applications of ODEs

ODEs

celestial motion - <https://www.youtube.com/watch?v=kUXskc76ud8>

Neuroscience models - https://www.youtube.com/watch?v=bo1cYjcjV-Y&ab_channel=BlueBrainProject-EPFL

AI algorithms

robots - https://www.youtube.com/watch?v=WOwLquiFbPE&ab_channel=AntoninRaffin

PDEs

climate simulation - https://www.youtube.com/watch?v=f-mp20H2nVo&ab_channel=NCARVisLab

NCAR research - <https://www.youtube.com/c/NCARVisLab>

NCAR derecho - https://arc.ucar.edu/knowledge_base/74317833

Fluid dynamics

wind farm - <https://www.youtube.com/watch?v=NHNL5JRQnHQ>

heart valve - https://www.youtube.com/watch?v=l9NITV72lnQ&ab_channel=RiceUniversity

https://www.youtube.com/watch?v=to9mhtxOff8&ab_channel=MiguelA.Fernandez

microfluidics - https://www.youtube.com/watch?v=ZCMr1tHnGRs&ab_channel=FLOW-3D

jelly fish - <https://www.youtube.com/watch?v=oPTPHDe0EPM>

black hole - https://www.youtube.com/watch?v=i2u-7LMhwwE&ab_channel=NASAGoddard